

Your Professional City Finder

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Abstract—Choosing a suitable city for independent living has become increasingly difficult for young adults due to rising housing costs, income disparities and differences in quality of life. Existing solutions are often based on static rankings that provide limited support for personalized exploration and decision-making. This project presents an interactive visual analytics system designed to support the exploration and comparison of cities using multiple socio-economic indicators. The system integrates heterogeneous datasets, computes a customizable score for each city and provides coordinated visualizations including a world map, ranking views and city comparison panels. Users can dynamically adjust attribute weights, compare multiple cities and explore detailed information through interactive visualizations. The resulting system demonstrates how visualization techniques can transform complex multidimensional datasets into interactive insights for urban decision-making.

Index Terms—Information Visualization, Visual Analytics, Interactive Visualization, Dashboard Design, Urban Data, Decision Support

I. INTRODUCTION

Achieving financial and residential independence has become increasingly challenging for young adults. Rising housing costs, differences in average salaries and significant variations in quality of life make it difficult to determine which cities provide the best opportunities for independent living.

While many online platforms such as Numbeo [1] provide rankings and statistics about cities, these solutions are typically static and do not allow users to explore trade-offs between multiple factors simultaneously. Furthermore, different individuals may prioritize different aspects when choosing a city. Some users may prioritize affordability, while others may value income potential or overall quality of life.

Therefore, the goal of this project is to design and implement an interactive visual analytics system that supports city comparison and decision-making. The proposed system integrates multiple datasets into a unified framework and provides interactive visualizations that allow users to compare cities, adjust priorities and gain insight into complex urban indicators, enabling users to explore multidimensional datasets, identify patterns and compare alternatives interactively.

II. DATASET AND DATA PROCESSING

The system integrates information from multiple public sources related to independent living. Eight datasets were collected from Kaggle [2], covering indicators associated with cost of living, average salaries, quality of life and other urban characteristics relevant to young adults: Top Cities Quality of Life Index 2024, City Quality of Life Dataset, Global Cost of

Living, City Happiness Index 2024, Internet Download Speed 2024, Sunshine Hours for Cities, World Cities Average Temperature and World Cities Information. All datasets correspond to the 2023/24 period. The data integration process follows the classical visualization pipeline [3], [4] consisting of data collection, cleaning, transformation, mapping and rendering. Data originating from different sources required preprocessing before visualization. The preprocessing stage consisted of several steps:

- Merging heterogeneous datasets into a unified table.
- Removing duplicates and inconsistent entries.
- Resolving naming differences between cities.
- Handling missing values.
- Normalizing numerical attributes to comparable ranges.

Since the datasets originated from different sources and used different scales, normalization was necessary to ensure that attributes could be combined meaningfully when computing the final score. After preprocessing, the resulting dataset contains 138 cities described by 19 attributes.

III. SYSTEM DESIGN

A. Design Requirements

The system was designed following common information visualization principles [5]:

- Overview first
- Zoom and filter
- Details on demand

The objective was to provide users with an overview of all available cities while also allowing detailed inspection and comparison of specific locations.

Several design requirements were identified:

- Support comparison of multiple cities.
- Enable personalized weighting of attributes.
- Provide geographical context.
- Allow rapid exploration of large datasets.
- Present detailed information without overwhelming the user.

B. City Score

To simplify the comparison process, the project introduces the city score, a composite metric that combines multiple indicators into a single score.

Each attribute is normalized and combined using user-defined weights:

$$Score = \frac{\sum_{i=1}^n w_i a_i}{\sum_{i=1}^n w_i} \quad (1)$$

where:

- a_i represents a normalized attribute value.
- w_i represents the user-defined weight assigned to that attribute.

This approach (1) allows users to express personal preferences and instantly observe how rankings change when priorities are modified.

Rather than producing a fixed ranking, the city score supports exploratory analysis by enabling dynamic score recomputation.

C. Coordinated Views

The application is composed of multiple linked visualizations that work together to support exploration and comparison.

1) *Interactive World Map*: The map provides the primary overview of the dataset. Cities are represented as circles positioned according to their geographic coordinates.

Circle size is determined by the computed score, allowing users to quickly identify highly ranked cities while maintaining geographical context, as shown in Fig. 1.

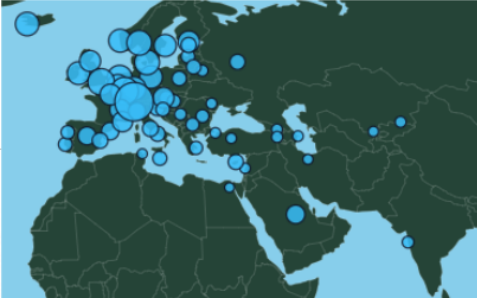


Fig. 1. Overview of the interactive world map.

2) *Comparison View*: An attribute comparison panel in Fig. 2 presents cities in bar charts according to their current city score in each attribute. This view provides a direct comparison between cities and allows users to quickly identify top-performing locations.

3) *City Card and Detailed Comparison*: The City Card serves as the primary details-on-demand component of the system. Each City Card combines both textual and graphical representations of the selected city. Textual information includes the city name, overall score and the exact values of relevant attributes, which can be seen in Fig. 3.

Individual attributes are presented using compact score indicators inspired by dashboard-style interfaces, as shown in Fig. 4. These visual summaries allow users to quickly identify strengths and weaknesses without scanning large tables of numerical values. The traffic-light color scheme further supports rapid interpretation by highlighting favorable, intermediate and unfavorable values.

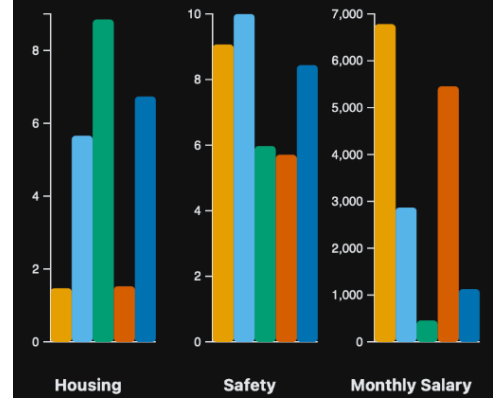


Fig. 2. Overview of the attribute comparison panel.

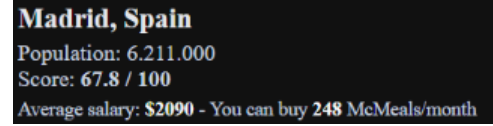


Fig. 3. Primary city general information.

A radar chart is integrated within the City Card to support multivariate comparison, as shown in Fig. 5. Each axis represents one of the selected attributes, while the resulting polygon provides a visual profile of the city. This representation is particularly useful because it highlights overall patterns that may be difficult to detect when comparing numerical values individually.

Together, these components transform the City Card into a coordinated analytical view that supports both detailed inspection and decision-making.

4) *Radar Chart*: The radar chart enables multivariate comparison between cities. Each axis represents an attribute and the resulting polygon provides a visual summary of city strengths and weaknesses. The radar chart is particularly useful when comparing multiple cities simultaneously, as shown in Fig. 7.

IV. INTERACTION DESIGN

Interaction plays a central role in the system. Rather than presenting static information, users actively explore the dataset through multiple interaction techniques.

A. City Selection

Cities can be selected directly from the map or from the ranking view. Selecting a city updates all coordinated views and highlights the corresponding information throughout the interface.

B. Attribute Selection

Users can choose which attributes should influence the analysis. This allows the exploration process to focus on factors that are most relevant for a particular decision-making scenario.

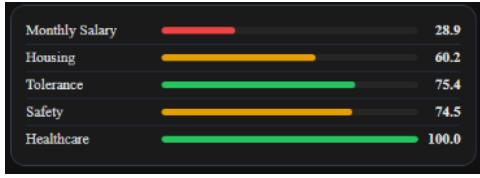


Fig. 4. Primary city scores displayed with traffic-light color scheme.

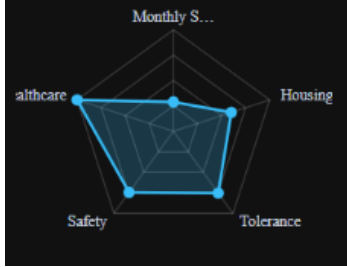


Fig. 5. Primary city radar chart with selected attributes.

C. Weight Adjustment

One of the main features of the system is the ability to modify attribute weights interactively, Fig. ??.

Slider controls allow users to increase or decrease the importance of specific indicators. The city score is automatically recalculated after each modification.

D. Hover Interactions

Hover interactions provide detailed information without introducing visual clutter. When users place the cursor over cities or graphical elements, additional information becomes available through tooltips, such as as label names or descriptions. This interaction technique supports details-on-demand while preserving a clean interface.

E. Coordinated Updates

All visual components are linked together. Any interaction performed in one view automatically updates the remaining views, allowing users to maintain context throughout the exploration process. This coordinated-view approach improves the overall analytical workflow and reduces cognitive effort.

V. VISUAL DESIGN DECISIONS

A. Color Design

Color was used carefully to support interpretation and comparison. For score indicators, a traffic-light color scheme was selected, where:

- Green indicates favorable values.
- Orange indicates intermediate values.
- Red indicates less favorable values.

This mapping is commonly used in dashboards and decision-support systems and can therefore be interpreted rapidly by users.

For city comparison, distinct categorical colors were selected to maximize perceptual separation and reduce confusion

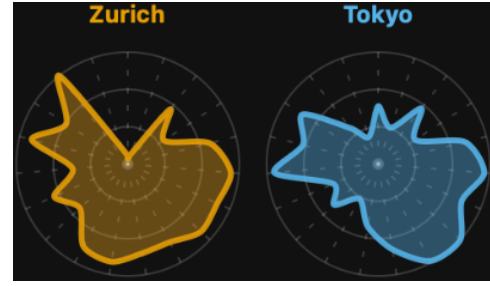


Fig. 6. Radar chart used for multivariate city comparison.



Fig. 7. Interactive weight adjustment controls.

when multiple cities are displayed simultaneously. Special attention was given to accessibility and color-blind friendliness during the design process.

B. Visual Encodings

Different visual variables were selected according to the nature of the represented data, as shown in Table I. The selected encodings follow established visualization principles and facilitate rapid interpretation.

TABLE I
VISUAL ENCODINGS USED IN THE SYSTEM

Variable	Visual Encoding
Score	Position and Length
City Identity	Color
Geographic Location	Position
Attribute Value	Radar Distance
Weight Importance	Slider Position

C. Perceptual Considerations

The design was informed by principles of human visual perception [6]. Color is a preattentive visual feature, allowing users to identify important information rapidly without conscious effort. High contrast was used to improve readability and visual hierarchy was applied to guide attention toward the most relevant components. Hence, color-blind friendly color palettes such as Okabe-Ito Palette, Fig. 8, were introduced in the color selection process.

Furthermore, the interface was also designed to minimize clutter while preserving sufficient detail for analytical tasks.

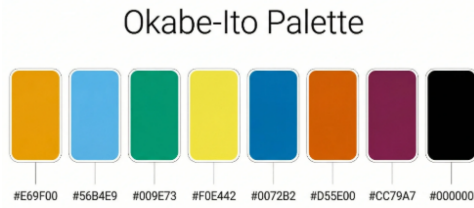


Fig. 8. Okabe-Ito color-blind accessible palette.

VI. EVALUATION

Due to project time constraints, a formal user study was not conducted. Instead, the system was evaluated through iterative testing during development and informal feedback from peers. This process helped identify usability issues and improve interaction mechanisms.

The evaluation focused primarily on three aspects:

- Functionality
- Usability
- Effectiveness

From a functional standpoint, dynamic score recalculation, weight adjustment, city filtering and linked view updates all performed as intended. The overall interaction design was also verified against Shneiderman’s visual information-seeking mantra [3], confirming that the interface consistently follows the overview-first, zoom-and-filter, details-on-demand principle. The coordinated views, dynamic weighting mechanism and comparison features were found to be intuitive and useful during exploratory analysis.

VII. DISCUSSION

The developed system demonstrates how visualization techniques can support urban decision-making by transforming heterogeneous datasets into an interactive analytical environment.

One of the main strengths of the system is its flexibility. Users are not restricted to a predefined ranking and can adapt the analysis according to personal preferences. The coordinated-view design also provides a balance between overview and detail, allowing users to navigate between global patterns and local information efficiently.

However, several limitations remain. First, the quality of the results depends on the quality and completeness of the underlying datasets. Missing values and inconsistencies may affect the final score.

Second, the city score relies on user-defined weights. Different weighting strategies may produce different rankings, making the resulting score inherently subjective.

Finally, the total number of supported cities is currently limited by the available datasets.

VIII. FUTURE WORK

Future work could expand the number of supported cities and integrate additional socio-economic indicators.

Another interesting direction would be the incorporation of real-time data sources, allowing the system to reflect current market conditions.

A formal user study involving young adults could also provide valuable insights regarding usability, effectiveness and decision-making support.

Finally, more advanced recommendation techniques could be integrated to assist users in identifying cities that best match their preferences.

IX. CONCLUSION

This project presented an interactive visual analytics system designed to support city comparison and urban decision-making. By combining heterogeneous datasets with coordinated visualizations and user-controlled weighting mechanisms, the system enables the exploration of complex urban indicators in an intuitive and flexible manner.

The resulting application demonstrates how interactive visualization techniques can transform multidimensional datasets into meaningful insights and support informed decision-making for young adults seeking independent living opportunities.

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